Software Requirements Specification

For

‘SmartFarm’

Version 2.1

Prepared by

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SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

SmartFarm Management System

1.Introduction

1.1 Purpose

The purpose of the ***SmartFarm*** farming management system is to provide a centralized, intelligent platform that helps farmers and agricultural managers efficiently plan, monitor, and control all aspects of farm operations. The system is designed to replace manual, fragmented systems, and paper-based processes with a unified digital solution that improves decision-making, productivity, and overall farm performance.

At its core, the system aims to simplify the management of key farming activities by bringing together multiple functional areas such as inventory tracking, sales management, employee coordination, task scheduling, and environmental monitoring into a single, easy-to-use interface. This integration enables users to have a real-time overview of their farm's status and make informed decisions quickly.

The system also serves as a decision-support tool by providing data-driven insights. For example, it tracks revenue trends, identifies inventory shortages, monitors employee activity, and highlights urgent tasks. By presenting this information through dashboards and visual summaries, the system helps users identify problems early and take corrective action.

A key purpose of the SmartFarm system is to enhance agricultural intelligence through technology. It incorporates features such as:

- Weather monitoring and forecasting, which allows farmers to plan activities like irrigation, planting, and harvesting based on environmental conditions
- AI-assisted disease detection, which helps users diagnose crop or livestock issues and receive recommendations for treatment and prevention

These intelligent features reduce uncertainty and improve the accuracy of farming decisions.

Additionally, the system aims to improve resource management and cost efficiency. By tracking inventory levels, monitoring sales transactions, and managing payroll, the system helps minimize waste, prevent shortages, and ensure proper financial planning.

The SmartFarm Management System is also designed to support operational coordination by enabling task assignment and event scheduling. This ensures that farm workers know their responsibilities and deadlines, improving productivity and accountability.

Ultimately, the purpose of the SmartFarm Management System is to:

Digitize and streamline farm operations,
Improve efficiency and reduce manual workload,
Support informed decision-making through real-time data,
Enhance productivity and profitability,
Introduce smart, technology-driven farming practices.

By achieving these goals, the system empowers farmers to move from traditional farming methods to a more modern, data-driven approach to agriculture.

1.2 Scope

The ***SmartFarm*** management system is designed to cover the core operational, managerial, and decision-making activities involved in modern farming. Its scope includes the integration of multiple farm management functions into a single platform, enabling users to efficiently monitor, control, and optimize their agricultural operations.

The system provides a comprehensive digital environment where all key aspects of farm management are interconnected. Instead of using separate tools or manual records, users can manage their entire farm through one system, ensuring consistency, accuracy, and real-time access to information.

Within its scope, the system supports the following major areas:

Farm Monitoring (Dashboard)

The system provides a centralized dashboard that gives users a real-time overview of farm performance. It displays key indicators such as revenue, inventory status, employee activity, and pending tasks. This allows farm managers to quickly assess the overall condition of operations and make timely decisions.

Inventory Tracking

SmartFarm manages all farm inputs and resources, including seeds, fertilizers, chemicals, equipment, and harvested products. It tracks stock levels, monitors minimum thresholds, and generates alerts when items are running low. This helps prevent shortages, reduce waste, and ensure smooth farm operations.

Sales and Revenue Management

The system records and tracks all sales transactions, including product type, quantity, buyer details, and payment status. It calculates total revenue, distinguishes between completed and pending sales, and provides insights into financial performance. This supports better financial planning and profitability analysis.

Employee and Payroll Management

SmartFarm maintains records of all farm workers, including their roles, salaries, and employment status. It also calculates payroll, including deductions and net pay. This ensures accurate salary management and improves workforce organization and accountability.

Task Scheduling and Event Planning

The system allows users to create, assign, and monitor tasks across the farm. Tasks can be prioritized, scheduled, and linked to specific locations or employees. Additionally, events such as deliveries, meetings, or farm activities can be planned and tracked a calendar system. This improves coordination and ensures that operations are completed on time.

Weather Monitoring and Forecasting

The system integrates external weather data to provide real-time environmental conditions and short-term forecasts. This information helps farmers plan activities such as planting, irrigation, spraying, and harvesting. It also includes farming advisories based on weather conditions to reduce risks and improve yields.

AI-Based Crop and Livestock Disease Diagnosis

SmartFarm includes an intelligent assistant that helps users identify possible diseases based on observed symptoms. By analysing user input, the system provides likely diagnoses, recommended treatments, and prevention strategies. This feature supports early detection and reduces crop or livestock losses.

2.Overall Description

2.1 Product Perspective

SmartFarm is a **web-based system** built using:

Frontend: HTML, CSS, JavaScript
Backend: PHP
External APIs: Weather API (Open-Meteo)

Future and Possible integrations may include:

AI APIs
SMS/WhatsApp/Email notifications
IoT farm sensors

2.2 Product Features

The SmartFarm Management System is composed of several integrated modules, each designed to handle a specific aspect of farm operations. These modules work together to provide a complete, centralized solution for managing agricultural activities efficiently and effectively.

1. Dashboard

The Dashboard serves as the central control panel of the system. It provides users with a real-time overview of key farm activities and performance indicators.

Key features include:

- Display of total revenue and financial summaries
- Overview of inventory status, including shortage alerts
- Summary of active employees and workforce status
- List of tasks due for the current day
- Snapshot of recent sales transactions
- Quick access to other modules

The dashboard enables users to quickly assess the overall state of the farm and identify urgent issues that require attention.

2. Weather Monitoring

The Weather Monitoring module provides real-time weather data and forecasts to support farm planning and decision-making.

Key features include:

- Current weather conditions (temperature, humidity, wind speed, rainfall, etc.)
- Short-term weather forecasts (e.g., 7-day forecast)
- Integration with external weather APIs
- Farming advisory messages based on weather conditions

This module helps farmers plan activities such as irrigation, planting, spraying, and harvesting, reducing risks associated with unpredictable weather

3. AI Disease Detection

The AI Disease Detection module acts as an intelligent assistant that helps identify crop and livestock diseases.

Key features include:

User input of symptoms via text prompt
Analysis of symptoms using an inbuilt knowledge-based system
Identification of possible diseases
Recommendations for treatment and prevention
Quick reference and suggestions for common agricultural issues

This system supports early detection of diseases, helping to minimize losses and improve farm productivity.

4. Inventory Management

The Inventory Management module handles the tracking and control of all farm resources and inputs.

Key features include:

Adding, updating, and managing inventory items
Tracking quantities and units (e.g., kg, litters)
Setting minimum stock levels
Automatic generation of shortage alerts
Display of inventory status and total value
Supplier tracking

This system ensures efficient use of resources, prevents stock shortages, and supports proper planning of farm inputs.

5. Sales & Revenue Tracking

The Sales module manages all financial transactions related to farm products.

Key features include:

Recording sales transactions (product, quantity, price, buyer)
Tracking sales status (pending or completed)
Automatic calculation of total revenue
Display of sales history and summaries
Monitoring of financial performance

This system helps users track income, manage sales operations, and analyse profitability.

6. Employee & Payroll Management

This system manages farm workers and handles payroll calculations.

Key features include:

Adding and managing employee records
Storing employee details (role, salary, contact information)
Tracking employee status (active, on leave, etc.)
Payroll calculation, including deductions (e.g., NSSF, PAYE)
Display of total payroll and individual salaries

This system improves workforce management, ensures accurate salary processing, and enhances accountability.

7. Task & Event Management

The Task & Event Management module supports planning, scheduling, and coordination of farm activities.

Key features include:

Creation and assignment of tasks to employees
Setting task priorities and deadlines
Tracking task status (pending, in progress, completed)
Display of tasks due on specific dates
Calendar view for events and scheduled activities
Management of farm-related events (meetings, deliveries, etc.)

This module ensures that farm operations are well-organized and that tasks are completed efficiently and on time.

2.3 User Classes

User Type: Admin (System Administrator)

The admin is the highest-level user in the ***SmartFarm*** Management System and has full control over all system functionalities and data. This user is responsible for managing the overall operation of the system, ensuring that all modules function correctly, and maintaining system integrity.

The admin can access and manage all systems, including dashboard monitoring, inventory, sales, employees, tasks, weather data, and AI features. This role is typically assigned to the farm owner, system manager, or IT administrator.

Responsibilities of the Admin include:

- **System Management**

- Configure and maintain the system settings
    - Ensure all modules are functioning properly
    - Monitor overall system performance
- **User Management (if implemented)**
    - Create, update, and delete user accounts
    - Assign roles and permissions to users
    - Control access levels for different users
- **Data Management**
    - Add, edit, and delete records across all modules
    - Maintain accurate data for inventory, sales, employees, and tasks
    - Ensure data consistency and integrity
- **Operational Oversight**
    - Monitor farm activities through the dashboard
    - Review reports on revenue, inventory status, and workforce performance
    - Make strategic decisions based on system insights
- **Financial Control**
    - Oversee all sales transactions and revenue tracking
    - Approve and manage payroll processing
    - Ensure financial data is accurate and up-to-date
- **Task and Workforce Coordination**
    - Assign and monitor tasks
    - Supervise employee activities and performance
- **Security and Maintenance**
    - Ensure system security (e.g., access control, data protection)
    - Perform backups and system updates (if applicable)

2.4 Operating Environment

The ***SmartFarm*** Management System is designed to operate in a flexible and accessible environment, allowing users to interact with the system using commonly available devices and software platforms. The system is web-based, meaning it does not require installation on user devices and can be accessed through standard web browsers over the internet.

Web Browsers

The system is compatible with modern web browsers that support current web standards such as HTML5, CSS3, and JavaScript. Supported browsers include:

- Google Chrome
- Microsoft Edge
- Mozilla Firefox

These browsers must be updated to recent versions to ensure full compatibility, optimal performance, and security. The system interface is designed to render consistently across these browsers, providing a smooth and responsive user experience.

Devices

The SmartFarm system can be accessed from a variety of devices, making it convenient for users in different working environments:

- **Desktop Computers and Laptops**
  These provide the best experience for managing large amounts of data, such as viewing reports, handling inventory, and performing administrative tasks.
- **Mobile Devices (Smartphones and Tablets)**
  The system supports mobile access through responsive design, allowing users to:
  - Check dashboard updates in real time
  - View tasks and alerts while in the field
  - Enter data such as sales or inventory updates

This flexibility enables farm managers and workers to access the system both in the office and on-site.

Server Environment

The system runs on a web server that supports:

- PHP (for backend processing)
- A web server such as Apache or Nginx

- A database system (MySQL) for data storage

The server may be hosted:

- Locally (on-premise)
- On a remote/cloud hosting platform

3. Functional Requirements

3.1 Dashboard Module

FR-1: System shall display total revenue
FR-2: System shall display inventory alerts
FR-3: System shall display active employees
FR-4: System shall display today's tasks
FR-5: System shall show recent sales summary

3.2 Weather Module

FR-6: System shall fetch live weather data using API
FR-7: System shall display temperature, humidity, wind, and rainfall
FR-8: System shall provide 7-day forecast
FR-9: System shall generate farming advisory based on weather

3.3 AI Disease Detection Module

FR-10: User shall input symptoms via text
FR-11: System shall analyse symptoms
FR-12: System shall return diagnosis, treatment, and prevention
FR-13: System shall provide suggestions for common symptoms

3.4 Inventory Management

FR-14: User shall add inventory items
FR-15: System shall track quantity and minimum stock
FR-16: System shall generate shortage alerts
FR-17: System shall display inventory list
FR-18: System shall calculate total inventory value

3.5 Sales Module

FR-19: User shall record sales transactions
FR-20: System shall calculate total revenue
FR-21: System shall track pending vs completed sales
FR-22: System shall display sales history

3.6 Employee & Payroll Module

FR-23: User shall add employees
FR-24: System shall store salary and role
FR-25: System shall calculate payroll
FR-26: System shall calculate deductions (NSSF, PAYE)
FR-27: System shall display employee list

3.7 Tasks & Events Module

FR-28: User shall create tasks
FR-29: System shall assign tasks to employees
FR-30: System shall track task status
FR-31: System shall display tasks due today
FR-32: System shall manage events and calendar

4. Non-Functional Requirements

4.1 Performance

- System shall load pages within 2 seconds
- Weather updates every 10 minutes

4.2 Security

- Input validation required
- User authentication (future)
- Protection against XSS and CSRF

4.3 Usability

- Clean and intuitive UI
- Mobile-friendly design

4.4 Reliability

- System uptime ≥ 95%
- Backup support (future)

4.5 Scalability

- Support increasing number of farms and users

5.External Interface Requirements

5.1 User Interface

- Web-based dashboard with sidebar navigation

5.2 API Interface

- Weather API (Open-Meteo)

5.3 Hardware Interface (Future)

- IoT sensors (soil moisture, temperature)

6. Data Requirements

The ***SmartFarm*** Management System relies on structured data to support its operations, decision-making, and reporting functions. The system stores, processes, and retrieves data related to farm activities, resources, personnel, and transactions. Each data entity represents a key component of the system and is designed to ensure accuracy, consistency, and efficient data management.

1. Users

The Users entity represents individuals who access and interact with the system.

Key data includes:

User ID (unique identifier)
Full name
Username and password (for authentication)
Role (e.g., Admin, Manager, Worker)
Contact information (phone/email)
Account status (active/inactive)

2. Employees

The Employees entity stores information about farm workers and staff.

Key data includes:

- Employee ID
- Full name
- Role/position (e.g., supervisor, labourer)
- Salary
- Phone number
- Employment status (active, on leave, etc.)
- Start date

3. Inventory Items

The Inventory Items entity manages all farm resources and inputs.

Key data includes:

- Item ID
- Item name
- Category (e.g., seeds, fertilizer, equipment)
- Quantity available
- Unit of measurement (kg, litres, etc.)
- Minimum stock level
- Unit cost
- Supplier information

4. Sales Records

The Sales Records entity captures all transactions related to farm produce and products.

Key data includes:

- Sale ID
- Product name
- Quantity sold
- Unit price
- Total amount
- Buyer details
- Sale date
- Status (pending/completed)

5. Tasks

The Tasks entity manages work activities assigned within the farm.

Key data includes:

Task ID
Task title/description
Assigned employee
Location
Due date
Priority level (high, medium, low)
Status (pending, in progress, completed)

6. Events

The Events entity represents scheduled activities or important occurrences.

Key data includes:

Event ID
Event title
Description
Event type (e.g., meeting, delivery, inspection)
Date and time
Location

Data Relationships (Overview)

- A User can manage multiple Employees, Tasks, and Inventory Items
- An Employee can be assigned multiple Tasks
- Sales Records are linked to products from Inventory Items
- Tasks and Events may be associated with specific employees or locations

7. Assumptions & Constraints

Assumptions:

- Users have internet access
- Users have basic digital literacy

Constraints:

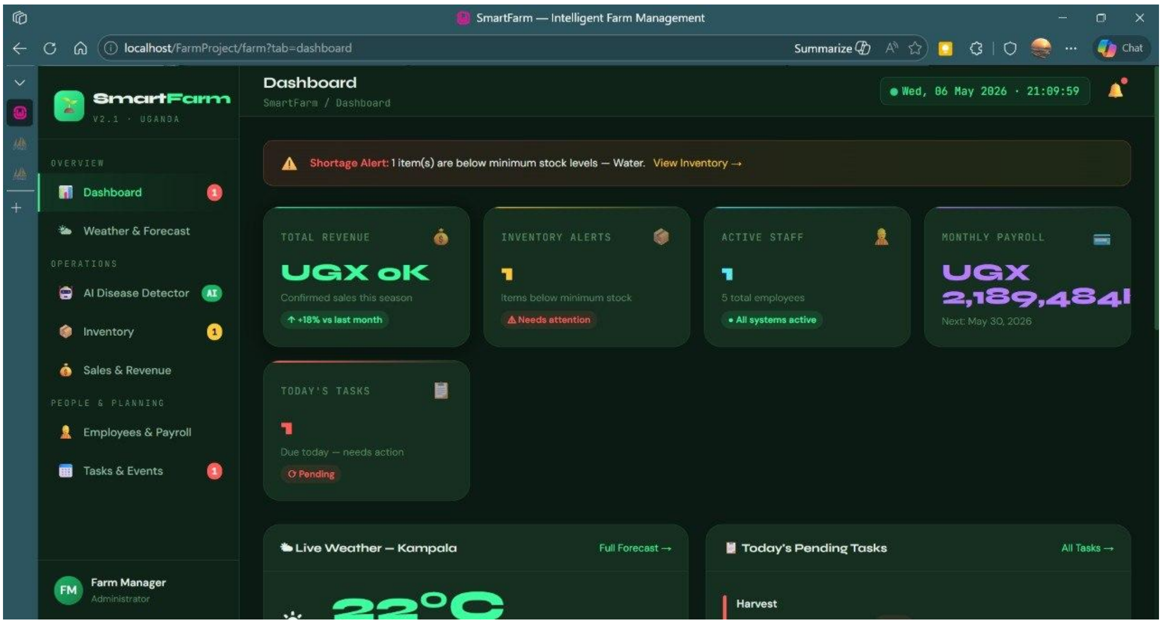
- Built using PHP stack
- Limited offline functionality
- AI system may not be very accurate

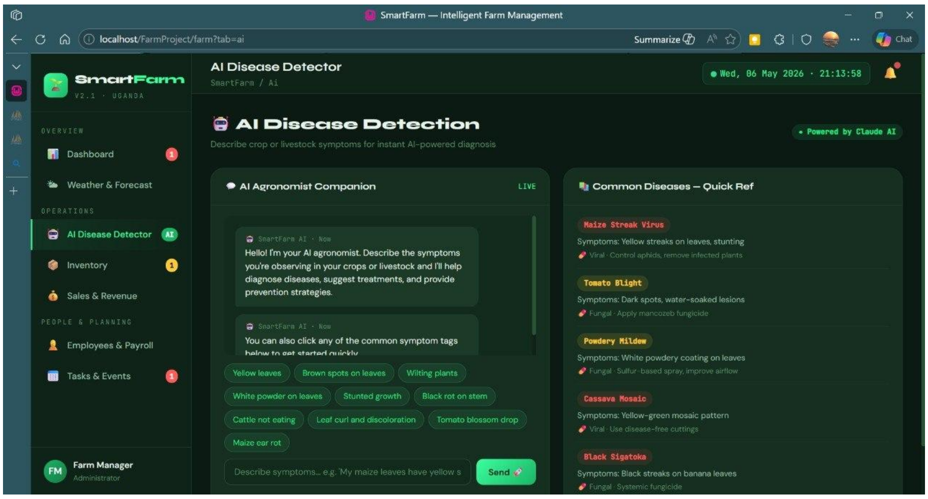
8. Future Enhancements

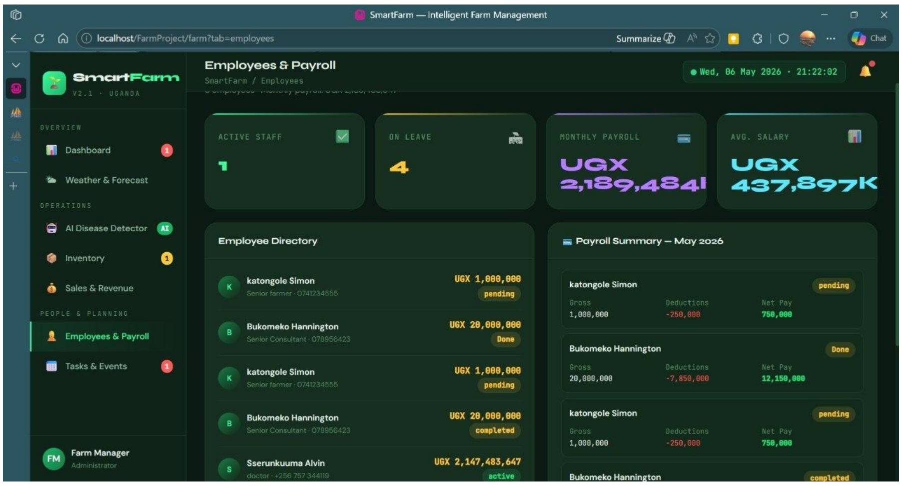
- Mobile app (Android)
- AI image-based disease detection
- SMS alerts
- Financial analytics & reports
- Multi-farm support

9. Appendices

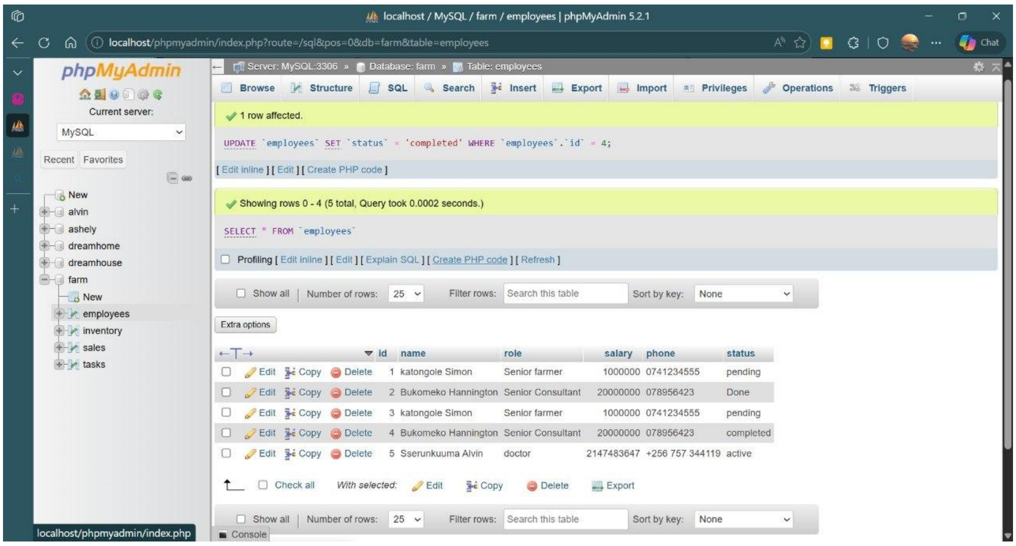
- A: UI screenshots

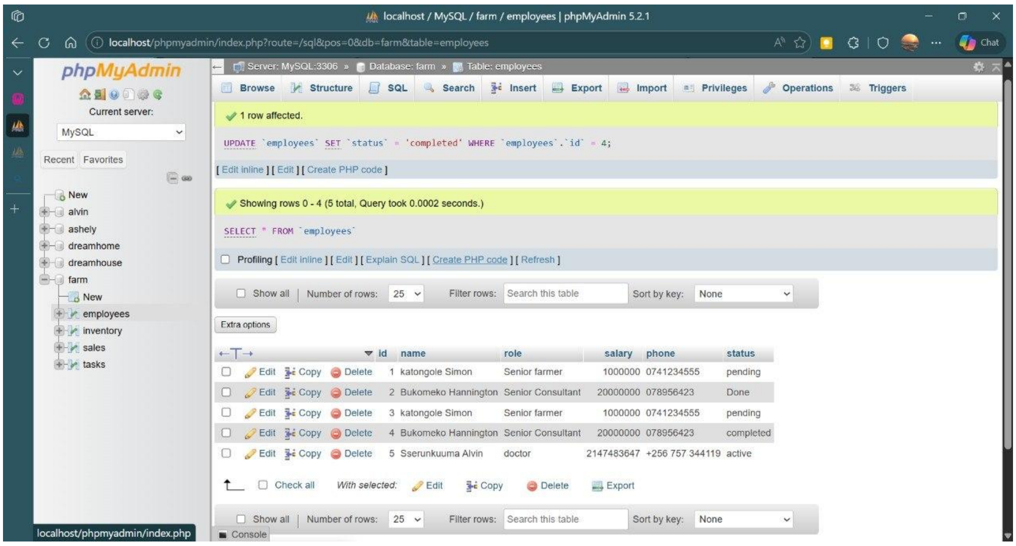






B: Database schema





C. Acronyms used

- **SRS**: Software Requirements Specification
- **PDO**: PHP Data Objects (Secure database access layer)
- **API**: Application Programming Interface
- **NSSF/PAYE**: Statutory tax and social security deductions in Uganda.

D. Open Meteo Weather Documentation

https://open-meteo.com/en/docs